

**Mid-Semester Examination (WS: 2023-24)
GPE201 (Earth and Planetary System)**

Date: 27 February 2024

Time: 14:00 – 16:00 hr

Maximum marks: 32

SOLUTION

Q.1

- During the Great Oxidation Event (GOE), wherein the oceans microbes evolved to produce oxygen via photosynthesis. The oxygen first accumulated in Earth's atmosphere at this time and has been present ever since.
- The tiny organisms known as cyanobacteria, or blue-green algae conducted photosynthesis using sun's radiation, water and carbon dioxide to produce carbohydrates and oxygen.
- Scientists estimated that 50-80% of the oxygen production on Earth comes from the ocean. The majority of this production is from oceanic plankton - drifting plants, algae, and some bacteria that can photosynthesize.
- Our planet Earth got oxygen sometime between 2.5 and 2.3 billion years ago.

Q.2

The International Astronomical Union (IAU) has established specific criteria for classifying celestial bodies as dwarf planets. According to these standards, a celestial body must meet the following criteria to be classified as a dwarf planet:

1. **Orbiting the Sun:** A dwarf planet must orbit the Sun directly and not be a satellite of another planet.
2. **Spherical shape & size:** It must have enough mass for its gravity to pull it into a nearly round (spherical) shape. This criterion distinguishes dwarf planets from smaller, irregularly shaped objects. Their diameters lie approximately between 1000 and 3000 km.
3. **Not clearing its orbit:** Typical planets have cleared their orbits of other debris, such as asteroids or smaller objects, due to their larger size and gravitational influence. Dwarf planets, on the other hand, have not cleared their orbits and often share their orbital zone with other objects.
4. **Location:** Typical planets are found in the inner or outer regions of a solar system, while dwarf planets are often located in the outer regions beyond the orbit of Neptune in a region known as the Kuiper Belt.

Examples of recognized dwarf planets include Ceres, Orcus, Pluto, Salacia, Haumea, Quaoar, Makemake, Gonggong, Eris, and Sedna. These bodies share characteristics with regular planets, such as being spherical due to their gravitational forces. However, they differ as they have not cleared their orbits of other objects, which is a key distinction according to the IAU's criteria. This characteristic places them in a category distinct from that of typical planets.

Q.3

Mid-ocean ridges are long, narrow, elevations of the seafloor with steep sides and irregular topography. They are spreading centres creating new oceanic crust at diverging plates. These are the most conspicuous features of the ocean basins and are continuous chain of mountains of young, basaltic rock at an active spreading center of the world ocean.

- i. It runs from the central Arctic Ocean down through the Atlantic, through the Indian Ocean and northward through the eastern Pacific.
- ii. The mid-ocean ridge is approximately 1000 km wide and rises 1000 to 3000 m above the adjacent seafloor.
- iii. The mid-ocean ridge is about 70,000 km long encircling the globe. It covers 21% of Earth's surface.

- iv. Many portions of the ridge have valleys that run along the crest of the ridge. These are called central or axial rift valleys.

The ocean ridge areas are the youngest parts of the seafloor and thus there hasn't been time for much sediment to accumulate.

Q.4

To calculate the circumference of the Sun, we can use the concept of angular diameter. The angular diameter of an object is the angle it subtends at an observer's eye. We can use the formula:

Given that the Sun's angular diameter is approximately 30 arcminutes (0.5 degrees) and its distance from Earth is 1.50×10^8 km, we can calculate the diameter of the Sun using:

$$\text{Diameter} = \text{Distance} \times \tan(\text{Angular Diameter})$$

Converting the angle to radian:

$$\text{Angle (rad)} = (30 \times \pi) / (180 \times 60) = \pi / 360$$

$$\text{Diameter} = 1.50 \times 10^8 \times \tan(\pi / 360)$$

$$\text{Diameter} \approx 1.50 \times 10^8 \times 0.0087266$$

$$\text{Diameter} \approx 1.31 \times 10^6 \text{ km}$$

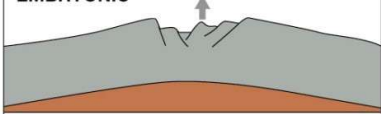
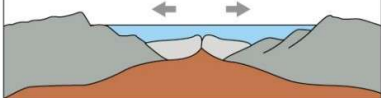
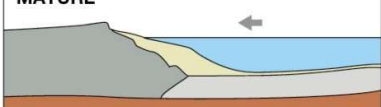
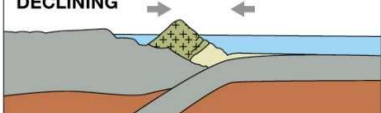
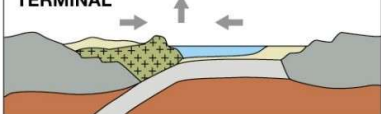
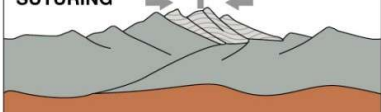
The circumference of the Sun is given by:

$$\text{Circumference} = \pi \times \text{Diameter}$$

$$\text{Circumference} \approx \pi \times 1.31 \times 10^6$$

$$\text{Circumference} \approx 4.12 \times 10^6 \text{ km}$$

Q.5

STAGE	MOTION	PHYSIOGRAPHY	EXAMPLE
EMBRYONIC 	Uplift	Complex system of linear rift valleys on continent	East African rift valleys
JUVENILE 	Divergence (spreading)	Narrow seas with matching coasts	Red Sea
MATURE 	Divergence (spreading)	Ocean basin with continental margins	Atlantic and Arctic Oceans
DECLINING 	Convergence (subduction)	Island arcs and trenches around basin edge	Pacific Ocean
TERMINAL 	Convergence (collision) and uplift	Narrow, irregular seas with young mountains	Mediterranean Sea
SUTURING 	Convergence and uplift	Young to mature mountain belts	Himalaya Mountains

Q.6

German astronomer Alfred Wegner proposed the theory of continental drift in 1912 with the following evidences:

- Fit of continental coastlines
- Different continent have nearly identical fossils
- Continuity of structures and formations
- Widespread Paleozoic glaciation in southern continent

Wegener's hypothesis lacked a geological mechanism to explain how the continents could drift across the earth's surface, and the prevailing belief that the earth was solid, resulted in the scientific dismissal of his theories.

Q.7

Moonquakes and Marsquakes are caused by different mechanisms compared to earthquakes on Earth, where plate tectonics are a dominant factor. **Moonquakes** can be triggered by several factors:

- (a) **Tidal Forces:** Tidal forces from Earth can cause stress on the Moon's surface, leading to moonquakes. Just as Earth's tides are caused by the gravitational pull of the Moon, the Moon experiences tidal forces from Earth that can cause its surface to flex and deform, generating seismic activity.
- (b) **Impact Events:** The Moon is heavily cratered due to impacts from asteroids and meteoroids. Large impacts can generate seismic waves that reverberate through the Moon, causing moonquakes.
- (c) **Lunar volcanism:** While most volcanic activity on the Moon ceased billions of years ago, some relatively recent volcanic activity has been observed. The movement of magma beneath the surface can also generate seismic waves.
- (d) **Internal processes:** The internal processes, such as the gradual cooling and contraction of the Moon's interior, can cause the crust to crack and generate seismic activity.

These factors combine to make the Moon seismically active, although moonquakes are generally less frequent and less intense than earthquakes on the Earth.

Marsquakes, on the other hand, are primarily caused by:

- (a) **Impact events:** Mars, like the Moon, is heavily cratered due to impacts from asteroids and meteoroids. Large impacts can generate seismic waves that can be detected by seismometers on the planet's surface.
- (b) **Volcanic Activity:** Mars has experienced volcanic activity in the past, and there is evidence of relatively recent volcanic flows on the planet. The movement of magma beneath the surface can generate seismic activity.
- (c) **Landslides:** Mars has steep slopes and cliffs, and landslides can occur, especially in areas with loose, unstable material. These landslides can generate seismic waves.
- (d) **Thermal expansion and contraction:** Mars has polar ice caps that expand and contract with the seasons. It also experiences large temperature variations between day and night. This can cause the planet's surface rocks to expand and contract, leading to stress and potentially generating seismic activity.

While plate tectonics as seen on Earth are not the drivers of seismic activity on the Moon and Mars, these celestial bodies still experience significant geological processes that result in quakes.

Q.8

Minerals demonstrate their properties through color, streak, and luster.

Color is the visible appearance of a mineral, which can vary due to impurities or other factors.

Streak refers to the color of the powdered form of a mineral, often determined by rubbing it against a hard white surface.

Luster describes how light is reflected from the surface of a mineral, with terms like metallic, vitreous (glassy), or dull used to characterize it. These properties help identify and classify minerals.

Q.9

We are aware of the following relationships from the fundamentals of radioactivity:

$$N = N_0 e^{-\lambda t} \dots\dots\dots(1)$$

$$t_{1/2} = \ln 2 / \lambda \dots\dots\dots(2)$$

Where, N_0 represents the initial number of radioactive atoms, N is the number of atoms present after decay at any time t , $t_{1/2}$ represents the half-life, and λ is the decay constant.

Additionally, we know from the question that the half-life ($t_{1/2}$) of ^{14}C is 5730 years, and N_0 is 10^{-11} g of ^{14}C .

Since atomic mass of ^{14}C is 14, therefore molar mass of ^{14}C will be: 14 g/mole

Further in order to convert grams to moles, we can use the following relationship:

$$\text{Moles} = \text{Mass (g)} / \text{Molar mass (g/mol)}$$

Therefore, 10^{-11} g of ^{14}C will have 10^{-11} g / 14 g/mole

$$N_0 = 7.1429 \times 10^{-13} \text{ moles.}$$

Hence 7.1429×10^{-13} moles corresponds to

$$N_0 = 7.1429 \times 10^{-13} \text{ moles} \times 6.02214076 \times 10^{23} \text{ atoms/mol} \quad (\text{Using Avogadro's number})$$

$$N_0 = 4.3016 \times 10^{11} \text{ atoms}$$

Substituting the value of λ in equation (2), we get:

$$\lambda = \ln 2 / t_{1/2} = \ln 2 / 5730$$

$$\lambda = 1.2097 \times 10^{-4} \text{ yr}^{-1}$$

Further, the remaining atoms of ^{14}C after a continuous decay for 2000 years:

$$= (4.3016 \times 10^{11}) \times e^{-(1.2097 \times 10^{-4}) \times 2000}$$

$$= 3.377 \times 10^{11} \text{ atoms}$$

OR

$$= 10^{-11} \times e^{-(1.2097 \times 10^{-4} \times 2000)}$$

$$= 7.851 \times 10^{-12} \text{ g}$$

Similarly, the ^{14}C remains after a continuous decay for one million years

$$= (4.3016 \times 10^{11}) \times e^{-(1.2097 \times 10^{-4} \times 10^6)}$$

$$= 1.25 \times 10^{-41} \text{ atoms}$$

OR

$$= 10^{-11} \times e^{-(1.2097 \times 10^{-4} \times 10^6)}$$

$$= 2.9067 \times 10^{-64} \text{ g}$$

Significance:

The above results indicate that after one million years, only 2.9067×10^{-64} g of ^{14}C , or approximately 1.25×10^{-41} atoms of ^{14}C , will remain. These amounts are extremely small, practically negligible. In other words, there will be no detectable atoms left. Therefore, when using radioactivity for dating purposes, it's crucial to select element with appropriate half-life.

Q.10

Bode's law: In 1772 the German astronomer Johann Bode devised an empirical formula to express the approximate distances of the planets from the Sun in Astronomical Unit.

A series of numbers is created in the following way: the first number is zero, the second is 0.3, and the rest are obtained by doubling the previous number. This gives the sequence 0, 0.3, 0.6, 1.2, 2.4, 4.8, 9.6, 19.2, 38.4, 76.8, etc. Each number is then augmented by 0.4 to give the sequence: 0.4, 0.7, 1.0, 1.6, 2.8, 5.2, 10.0, 19.6, 38.8, 77.2, etc. This series can be expressed mathematically as follows:

$$dn = 0.4 \text{ for } n = 1$$

$$dn = 0.4 + 0.3 \times 2^{(n-2)} \text{ for } n \text{ is more than and equal to } 2.$$

Distance between Sun and Jupiter (d6):

For Jupiter $n=6$

$$\text{Therefore } d_6 = 0.4 + 0.3 \times 2^{(6-2)}$$

$$= 0.4 + 0.3 \times 16$$

$$= 0.4 + 4.8$$

$$= 5.2 \text{ AU}$$

Astronomical Unit (AU) is a unit of measurement equal to 149.6 million kilometres, the mean distance from the centre of the earth to the centre of the sun.